Table 11.3 Analysis of two species population interactions

S. NO.	TYPES OF INTERACTION	SPECIES 1	SPECIES 2	GENERAL NATURE OF INTERACTION	EXAMPLES
1	Amensalism	–	0	The most powerful animal or large organisms inhibits the growth of other lower organisms	Animals destroyed at the feet of elephants
2	Mutualism	+	+	Interaction favorable to both and obligatory	Between crocodile and bird
3	Commensalism	+	0	Population 1, the commensal benefits, while 2 the host is not affected	Sucker fish on shark
4	Competition	–	–	Direct inhibition of each species by the other	Birds compete with squirrels for nuts and seeds
5	Parasitism	+	–	Population 1, the parasite, generally smaller than 2, the host	*Ascaris* and tapeworm in human digestive tract
6	Predation	+	–	Population 1, the predator, generally larger than 2, the prey	Lion predatory on deer

MUTUALISM (+, +): It is the type of interaction where both species benefit from the interaction. Mutualism may be facultative when the species involved are capable of existence independent of one another, or obligate, when the relationship is imperative of the existence of one or both species. Examples:

- Certain bacteria in the caeca and intestine of herbivores aid in the digestion of cellulose. In return the host provides suitable environment for the growth of the bacteria.
- The cross pollination of flowers by insects and birds seeking nectar and pollen which is of great importance in agriculture.
- Small birds cleaning the teeth of crocodiles. Here the birds gets food and the teeth of crocodile is cleaned.
- The hermit crab carries along on its shell a Sea anemone (a sedentary coelenterate). The crab is protected from its enemies by the stinging cells of the anemone while the anemone gets its food (**Fig. 11.12).**



Fig. 11.12 Sea anemone on hermit crab; Crocodile and bird

COMMENSALISM (+, 0): This defines the interaction in which two or more species are mutually associated in activities centering on food and one species at least, derives benefit from the association while the other associates are neither benefited nor harmed. The concept of commensalism has been broadened in recent years, to apply to coactions other than those centering on food such as cover, support, production, and locomotion. Examples:

- Barnacles attached to Whales travel thousands of miles collecting and filtering food from the moving water. The whales are not affected by the barnacles.

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